Outlook	Temperature	Humidity	Wind	Rain
Sunny	Hot	High	Weak	No
Sunny	Mild	High	Strong	No
cloudy	Hot	High	Weak	Yes
raing	Mild	High	Weak	Yes
raing	Cool	Normal	Weak	Yes
raing	Cool	Normal	Strong	No
Cloudy	Mild	Normal	Strong	Yes
Sunny	Cool	Normal	Weak	No

entrophy calculation of prime attribute (Rain)

$$-\frac{4}{8}\log_2\left(\frac{4}{8}\right) - \left(-\frac{4}{8}\log_2\left(\frac{4}{8}\right)\right)$$

$= 0$

entrophy for Outlook

Sunny: $\frac{0}{3}\log_2\left(\frac{0}{3}\right) - \frac{3}{3}\log_2\left(\frac{3}{3}\right) = 0(\log_2(0)) - 1(\log_2(1)) = 0$

Cloudy: $\frac{2}{2}\log_2\left(\frac{2}{2}\right) - \frac{0}{2}\log_2\left(\frac{0}{2}\right) = 1(\log_2(1)) - 0(\log_2(0)) = 0$

Rainy: $\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right) = -0.390 - (-0.528)$

$= 0.138$

$\frac{3}{8} \times 0 + \frac{2}{8} \times 0 + \frac{3}{8} \times 0.138$

$= 0.05175$

IG(Outlook) $= 0 - 0.05175$

$= -0.05175$

LUCKY

entrophy for temperature

Hot: $\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) = 0$

Cool: $\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right) = -0.528 - (-0.39) = -0.138$

Mild: $\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right) = -0.39 - (-0.528) = 0.138$

$$\frac{2}{8} \times 0 + \frac{3}{8} \times 0.138 + \frac{3}{8} \times 0.138$$

IG(Temperature) = 0

entrophy for Humidity:

Normal: $\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right) = 0$

High: $\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right) = 0$

$$\frac{4}{8} \times 0 + \frac{4}{8} \times 0$$

$$= 0$$

IG(Humidity) = 0 - 0 = 0

LUCKY

entropy for wind :

Weak : $\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right) = -0.442 - (-0.529) = 0.087$

Strong: $\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right) = (-0.528) - (-0.39) = -0.138$

$$\frac{5}{8} \times 0.087 + \frac{1}{3} \times 0.138 = 0.054 - 0.046$$

$$= 0.1$$

$$IG(Wind) = 0 - 0.1 = -0.1$$

We are choosing Humidity or Temperature as it has highest value (0)